QUANTUM STATE VISUALIZATION

DAY 23 | 30-DAY QUANTUM COMPUTING CHALLENGE

Experiment goal

Start with |0>, apply single-qubit gates, and visualize how the state changes on the Bloch sphere. Compare both state direction and measurement probabilities.

1. Why this project matters

A gate can change the measurement probabilities, the phase, or both. The Bloch sphere shows differences that a simple 0/1 histogram may hide. This experiment makes the effects of X, Y, Z and H visible.

2. State representation

A pure qubit is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$. Computational-basis measurement gives $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$.

Bloch vector: $x = 2\,\mathrm{Re}(\alpha^*\beta)$, $y = 2\,\mathrm{Im}(\alpha^*\beta)$, $z = |\alpha|^2 - |\beta|^2$. Here α^* denotes complex conjugation.

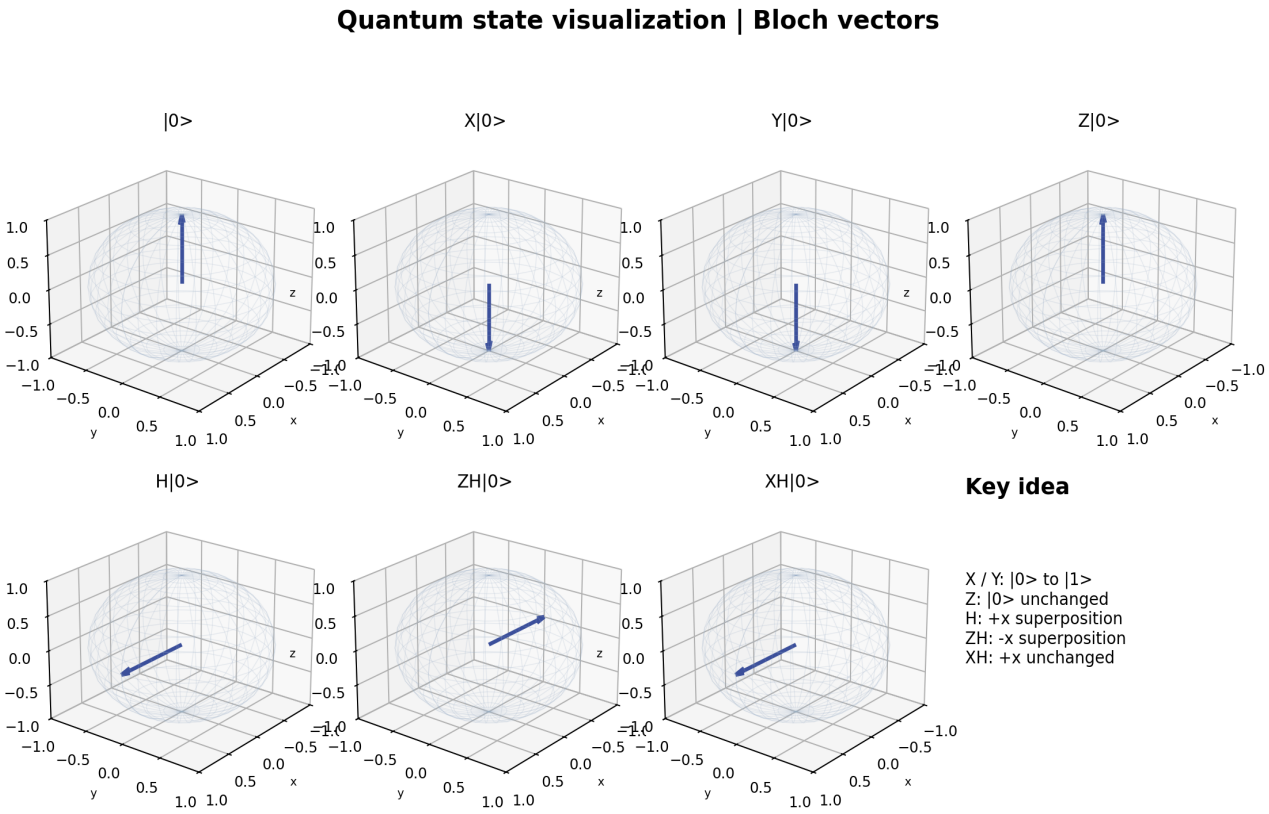
3. Gate effects from |0>

Gate	Result (up to global phase)	Bloch direction
Identity / Z	|0>	+z
X / Y	|1>	–z
H	(|0> + |1>)/√2	+x
Z after H	(|0> – |1>)/√2	–x
X after H	(|0> + |1>)/√2	+x

4. Circuit idea

For each experiment, prepare |0>, apply the selected gate or sequence, then calculate the statevector. Compare P(0), P(1) and the Bloch vector. No physical hardware is required.

5. Visualization: Bloch sphere



Each arrow represents a pure-state Bloch vector. X and Y move |0> to the south pole; Z leaves |0> at the north pole. H moves the state to the equator. Applying Z after H reverses the x direction.

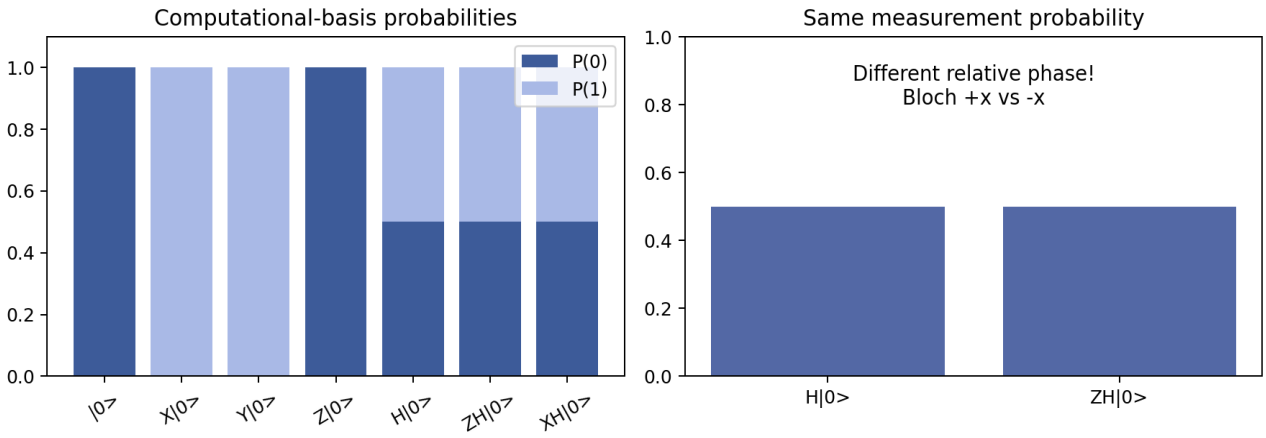
6. Why phase matters

H|0> = |+> and ZH|0> = |–>. Both have P(0) = P(1) = 0.5 when measured in the computational (Z) basis, but their relative phases differ. In the X basis they are perfectly distinguishable.

7. Expected observations

The Bloch directions are +z for |0>, –z for X|0>, +x for H|0>, and –x for ZH|0>. Y|0> differs from X|0> by a global phase only, which cannot be observed.

8. Measurement comparison



The same computational-basis probabilities can correspond to different quantum states. The chart compares measurement probabilities; the Bloch-sphere figure supplies the missing phase information.

9. Execution steps

• Install Python packages: **pip install numpy matplotlib notebook**.

• Open **quantum_state_visualization.ipynb** in Jupyter or VS Code.

• Run the cells in order to build gate matrices, calculate statevectors and plot Bloch vectors.

• Modify the **sequence** string to try new gate combinations; rerun the experiment.

• Save the figures and compare your observations with the expected states above.

• Optional: install Qiskit and execute the verification cell.

10. Interpretation and limitations

This is an ideal statevector experiment: no shot noise, gate errors or decoherence are included. Bloch spheres describe single-qubit states; multi-qubit entanglement needs additional tools. A gate does not always change computational-basis measurement statistics.

Summary: What to remember

X flips |0> to |1>. H prepares a balanced superposition. Z flips relative phase in a superposition. Measurement probabilities do not fully specify the state; visualizing its Bloch vector makes the difference clear.